
RESEARCH PAPER

Society Under Monetary Stress I: The Fiat Squeeze

Money Growth, Housing Affordability, and Household Formation

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Abstract

This paper asks whether the bridge from household income to housing access has become harder to cross. Public U.S. data from 1987 through 2024 show that home-price-to-income pressure rose substantially, while monthly-payment pressure moved with interest-rate regimes. State ACS/FHFA evidence from 2011 through 2024 links higher local housing-price pressure to higher rent burden, including in state-year panel models with state and year fixed effects. Evidence on under-35 household formation is weaker and proxy-dependent. The paper is descriptive, not causal, and is intended to help readers separate price pressure, payment pressure, and local rent stress when interpreting their own economic position.

JEL codes: D14, E31, E44, R21, R31 **Keywords:** housing affordability, monetary regimes, household formation, rent burden, public data **Disclosure:** This public-data descriptive study involves no human-subject intervention or randomized trial. Funding and conflict disclosures should be reviewed before any formal journal submission.

1. Introduction

Housing is where macroeconomics becomes personal.

A household does not experience money growth, inflation, credit conditions, or interest rates as abstract charts. It experiences them as a rent bill, a mortgage quote, a down payment target, a savings account that does or does not keep up, and the decision to buy, rent, move, wait, or stay with family longer than planned.

This distinction matters because a society can look wealthier in nominal terms while the path from labor income to durable ownership becomes harder. Wages can rise. Household income can improve. Asset values can climb. Yet if home prices and financing conditions move faster than ordinary income pathways, the household's practical agency may narrow even during periods that look prosperous in aggregate data.

This paper calls that pressure the **fiat squeeze**. The phrase is not meant as a slogan or a complete theory. It is a descriptive shorthand for a specific household experience: income may rise, credit may be available, and headline wealth may increase, but the conversion of income into durable ownership becomes more fragile. The squeeze is not just high prices. It is the interaction of prices, financing terms, saving capacity, and timing. A household that already owns an appreciating asset lives in a different regime than a household trying to buy into that same asset after the price level has shifted.

The paper asks a narrow empirical question: are public data consistent with a widening bridge between household income and housing access? It does not try to fully explain housing affordability. Zoning, construction costs, land constraints, taxes, credit standards, demographics,

and local labor markets all matter. The goal here is more modest: document the pattern, separate the price and payment channels, and test whether local housing-price pressure is associated with household stress indicators.

Four findings stand out.

First, national home-price-to-income pressure rose substantially. With 1987 set to 100, the national home-price-to-income index reached 151.0 by 2024. Second, monthly-payment pressure did not move in a straight line. Lower mortgage rates made payments easier through much of the 2010s even as prices rose. By 2024, however, the payment-to-income proxy had climbed back above its 1987 base, reaching 109.3. Third, the state-level extension strengthens the rent-burden channel: higher local housing-price pressure is associated with higher rent-burdened renter shares. Fourth, evidence on under-35 household formation is preliminary. The available proxies point only weakly in the expected direction and should not carry the paper's main claim.

The strongest conclusion is therefore careful but important: monetary and credit regimes shape the background conditions under which households try to turn income into durable security. The result is not immediate hardship every year. It is a longer, less reliable bridge from work to ownership.

2. Data

The national analysis uses public U.S. data, annualized where needed, from FRED, Census-linked income series, mortgage-rate data, and the Case-Shiller national home-price index. The core national sample runs from 1987 through 2024, bounded by the availability of the Case-Shiller home-price series and annual household income data.

The national variables include M2 money stock, the Consumer Price Index, the effective federal funds rate, the 30-year fixed mortgage rate, unemployment, the Case-Shiller national home-price index, nominal median household income, real median household income, the national homeownership rate, the personal saving rate, and real average hourly earnings where available.

The local extension uses FHFA state all-transactions House Price Index data and ACS 1-year state estimates. ACS years include 2011-2019 and 2021-2024; 2020 is omitted because the standard ACS 1-year release is unavailable. The merged ACS/FHFA state panel contains 663 state-year observations. Local outcomes include rent-burdened renter share, under-35 householder share, and ownership among under-35 householders.

The data choices are deliberately transparent. The paper favors measures that can be rebuilt from public sources and explained in plain language. That comes at a cost: the analysis is descriptive and cannot isolate causality. But it makes the empirical story easier to audit.

3. Measures

The national analysis uses four main measures.

The first is the **M2/CPI ratio index**, which compares money-stock growth to consumer-price growth, rebased to 1987 = 100. This is macro-regime context, not a single-cause explanation.

The second is the **home-price-to-income index**, which compares the national home-price index to nominal median household income, also rebased to 1987 = 100. This measures the price bridge from income to ownership.

The third is the **payment-to-income proxy**, which estimates the relative monthly principal-and-interest payment implied by the national home-price index and the 30-year mortgage rate. For each year, the proxy applies the standard fixed-rate mortgage amortization factor, $r(1+r)^{360} / ((1+r)^{360} - 1)$, where r is the monthly mortgage rate. That payment factor is multiplied by the national home-price index, rebased to the 1987 payment factor, and divided by nominal median household income. In plain language, the measure asks: if the home-price index and mortgage rate changed but the mortgage term stayed constant, how did the implied monthly payment move relative to income?

It is not an underwriting model. It does not include taxes, insurance, down payments, closing costs, credit scores, reserve requirements, or local price levels. Those omissions matter. A household can face a manageable monthly payment and still be blocked by the cash needed to buy, or by taxes, insurance, or underwriting constraints. The proxy is therefore a transparent index of the payment channel, not a complete affordability verdict. Higher values mean less payment affordability.

The fourth group covers household resilience: homeownership, personal saving, unemployment, and real income or wage context.

The local extension adds three state-level outcomes. Rent burden is the share of renter households paying at least 30 percent of income toward gross rent. Under-35 householdership is the number of owner and renter householders ages 15-24 and 25-34 divided by occupied housing units. Ownership among under-35 householders is the owner share within that under-35 householder group. These age-based measures are only proxies. They omit young adults who live in someone else's household and are not listed as householders.

The main local exposure is housing-price pressure, measured as $\log(\text{FHFA state HPI} / \text{ACS median household income})$. Alternative panel specifications use $\log$ FHFA HPI and $\log$ income separately.

4. National Results

4.1 The price bridge widened

Figure 1 compares the M2/CPI ratio, the home-price-to-income index, and the payment-to-income proxy. The M2/CPI ratio rose from 100.0 in 1987 to 275.1 in 2024. Over the same period, the home-price-to-income index rose to 151.0.

That does not mean money growth mechanically caused housing unaffordability. M2 is not an exogenous shock, and long-run indexed series can move together because of common trends. The better reading is more modest: the macro liquidity backdrop and the ownership-access hurdle changed substantially over the same period. For individual readers, the useful point is not a single-cause story; it is the need to watch whether income, asset prices, financing terms, and cash-entry requirements are moving in different directions.

The payment-to-income proxy followed a different path. It was below the 1987 base in 2000 and 2012 because lower mortgage rates offset some of the pressure from higher prices. By 2024, however, the proxy reached 109.3, reflecting the combined burden of higher home prices and higher mortgage rates.

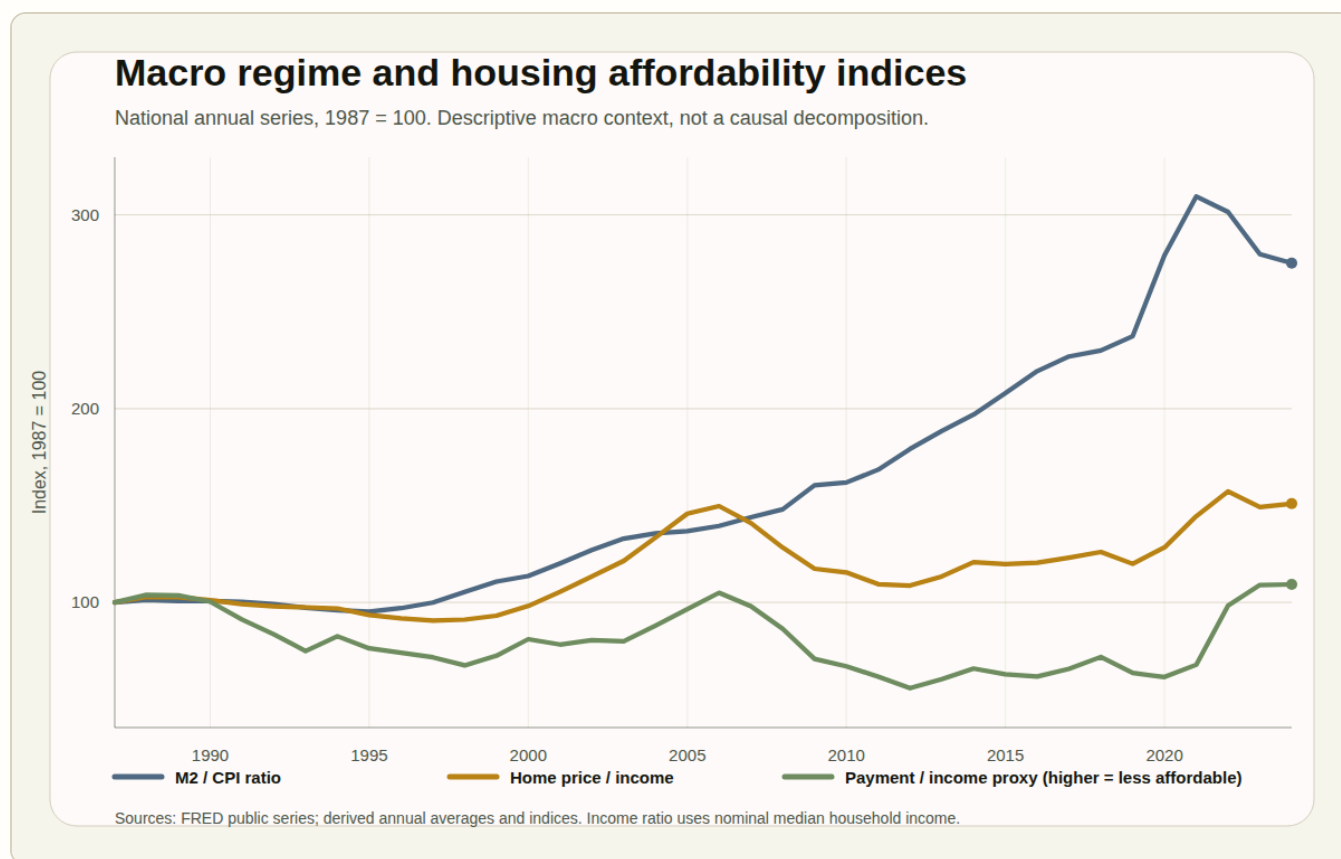


Figure 1. Macro regime and housing affordability indices

4.2 Rates changed the monthly-payment story

Figure 2 shows why affordability cannot be reduced to price alone. Mortgage rates were high in the late 1980s, fell across the 1990s and 2010s, and then rose again after 2021. This rate cycle shaped the monthly payment burden. Two households facing similar price-to-income ratios can face very

different monthly obligations depending on financing conditions.

The ex post real mortgage-rate proxy is included only as context. It is not a precise measure of each household's borrowing cost, and it is not the same as the inflation expectations embedded in a mortgage contract at origination. It simply shows how nominal mortgage rates compared with contemporaneous consumer-price inflation.

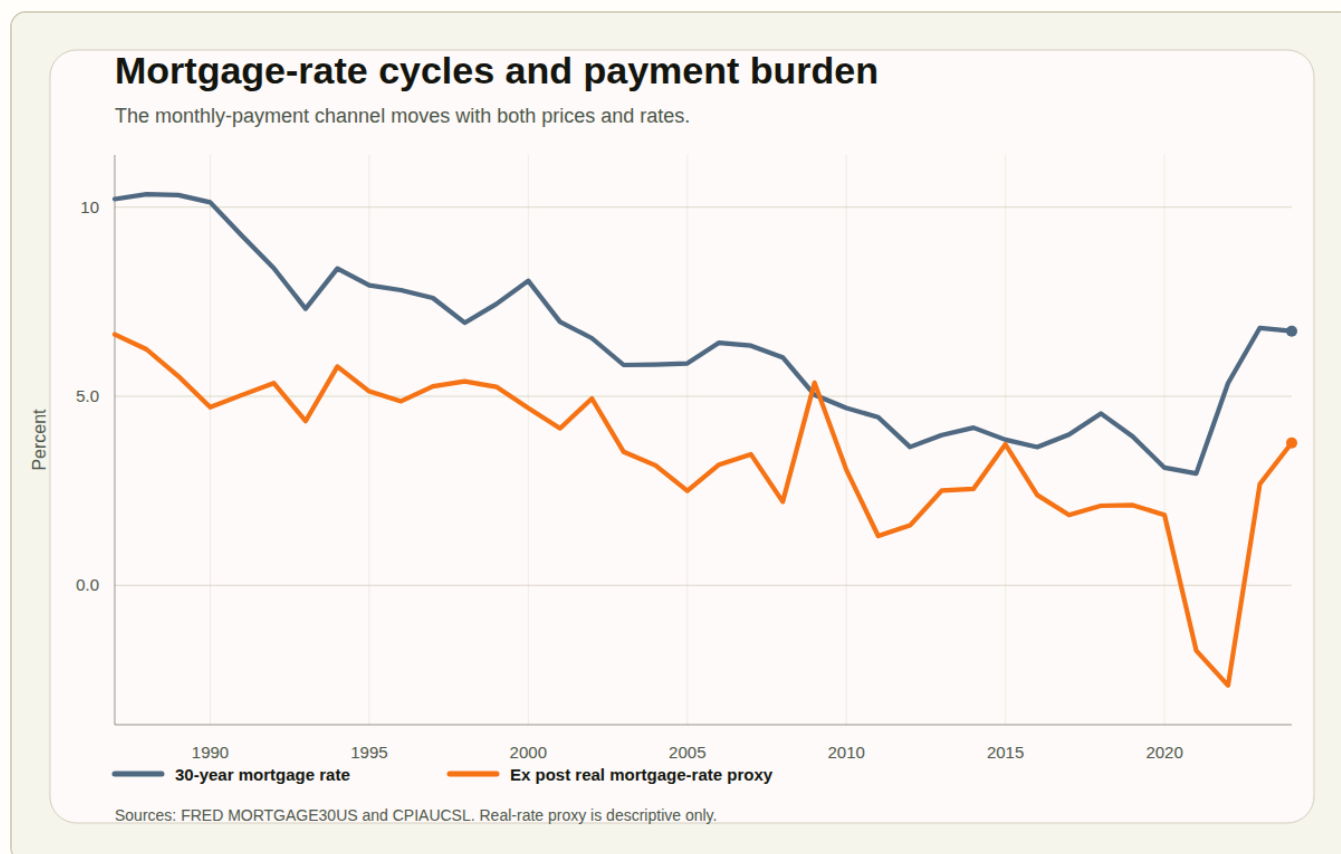


Figure 2. Mortgage-rate cycles and payment burden

4.3 Household resilience did not obviously offset the squeeze

Figure 3 indexes homeownership, the personal saving rate, and unemployment to 1987 = 100. These series do not move together. Homeownership rose into the mid-2000s, fell after the housing bust, and remained range-bound afterward. The saving rate moved through much larger swings, including the temporary 2020 spike. Unemployment captures the major labor-market stress episodes.

The point is not that these indicators prove household decline. They do not. The point is that they do not show an obvious household-resilience improvement large enough to erase the widening asset-access bridge.

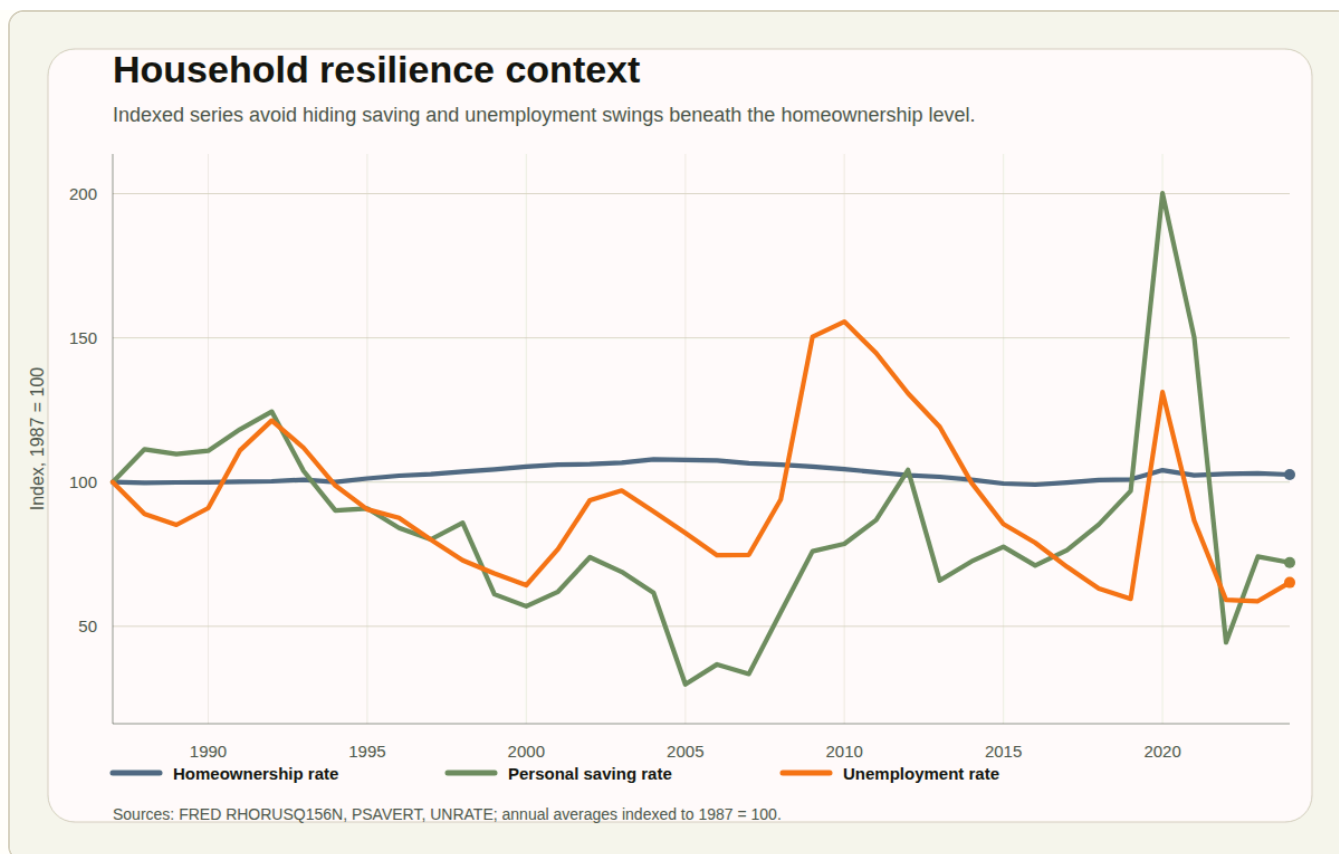


Figure 3. Household resilience context

4.4 Income improved, but the access hurdle rose faster

Figure 4 compares real median household income with the home-price-to-income index. Real income improved over the sample. That matters: the finding is not that households made no income progress. The finding is that income progress did not prevent the ownership-access bridge from widening.

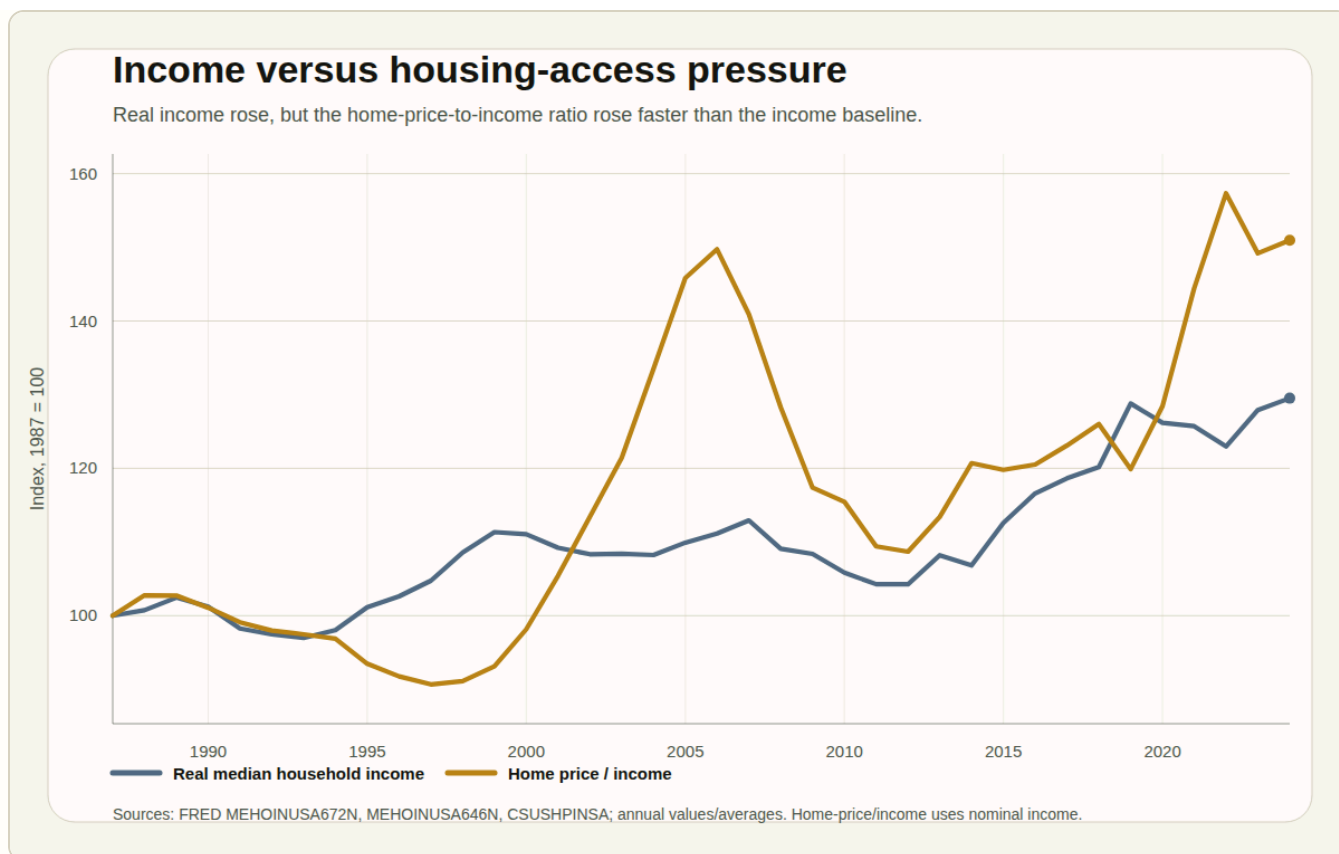


Figure 4. Income versus housing-access pressure

4.5 Milestone years

Table 1 shows why the story needs both the price channel and the payment channel. In 2007, home-price-to-income pressure was high, but the payment-to-income proxy remained below the 1987 base because mortgage rates were lower. In 2012 and 2019, low rates compressed the payment burden even while asset prices remained elevated. By 2024, both the home-price-to-income index and payment-to-income proxy were above their 1987 base.

Table 1. Milestone affordability and household-context indicators *Index columns use 1987 = 100. Rate columns are annual averages in percent.*

Year	M2/CPI ratio	Home price / income	Payment / income proxy	30-year mortgage rate	Ex post real mortgage-rate proxy	Homeownership rate	Personal saving rate
1987	100.0	100.0	100.0	10.21	6.64	63.98	7.54
2000	113.6	98.1	81.0	8.05	4.69	67.38	4.29
2007	143.9	140.9	98.0	6.34	3.47	68.15	2.52
2012	179.2	108.7	55.7	3.66	1.58	65.45	7.86
2019	237.4	119.9	63.6	3.94	2.12	64.55	7.31
2020	279.2	128.4	61.5	3.11	1.86	66.60	15.09
2024	275.1	151.0	109.3	6.72	3.77	65.62	5.43

From 1987 to 2024, the M2/CPI ratio rose by 175.1 percent, the home-price-to-income index rose by 51.0 percent, and the payment-to-income proxy rose by 9.3 percent. The smaller full-period change in payment burden hides the main regime story: payments became easier during the low-rate period, then tightened sharply when mortgage rates rose.

Simple correlations are useful only as descriptive signposts. The M2/CPI ratio is positively correlated with home-price-to-income pressure across annual observations ($r = 0.74$), but not with the payment-to-income proxy ($r = -0.16$). The 30-year mortgage rate, by contrast, is positively correlated with the payment-to-income proxy ($r = 0.67$). These correlations should not be read as causal estimates. Their practical value is to show that price pressure and payment pressure are related but distinct channels.

5. Local ACS/FHFA Extension

5.1 Rent burden is the clearest local channel

The ACS/FHFA extension adds a state-level household layer. It merges FHFA all-transactions state HPI data with ACS 1-year state estimates from 2011-2019 and 2021-2024. The endpoint exposure is nominal FHFA HPI growth from 2011 to 2024. The outcomes are changes in rent burden, under-35 householdership, and ownership among under-35 householders.

The strongest local endpoint result is rent burden. Across states and DC, HPI growth is positively correlated with the change in the rent-burdened renter share ($r = 0.43$). In plain English: states where home prices rose faster also tended to see larger increases in the share of renters paying at least 30 percent of income toward gross rent.

This relationship is descriptive and ecological. It does not prove that home-price growth caused rent burden to rise. But it is consistent with the paper's main affordability-pressure channel.

Robustness checks suggest the result is not only a top-growth-state artifact. The correlation remains positive after excluding the top three HPI-growth states ($r = 0.30$), after excluding the top five ($r = 0.26$), and after excluding both the top five and bottom five ($r = 0.32$). The relationship weakens outside the highest-growth states, but it does not disappear.

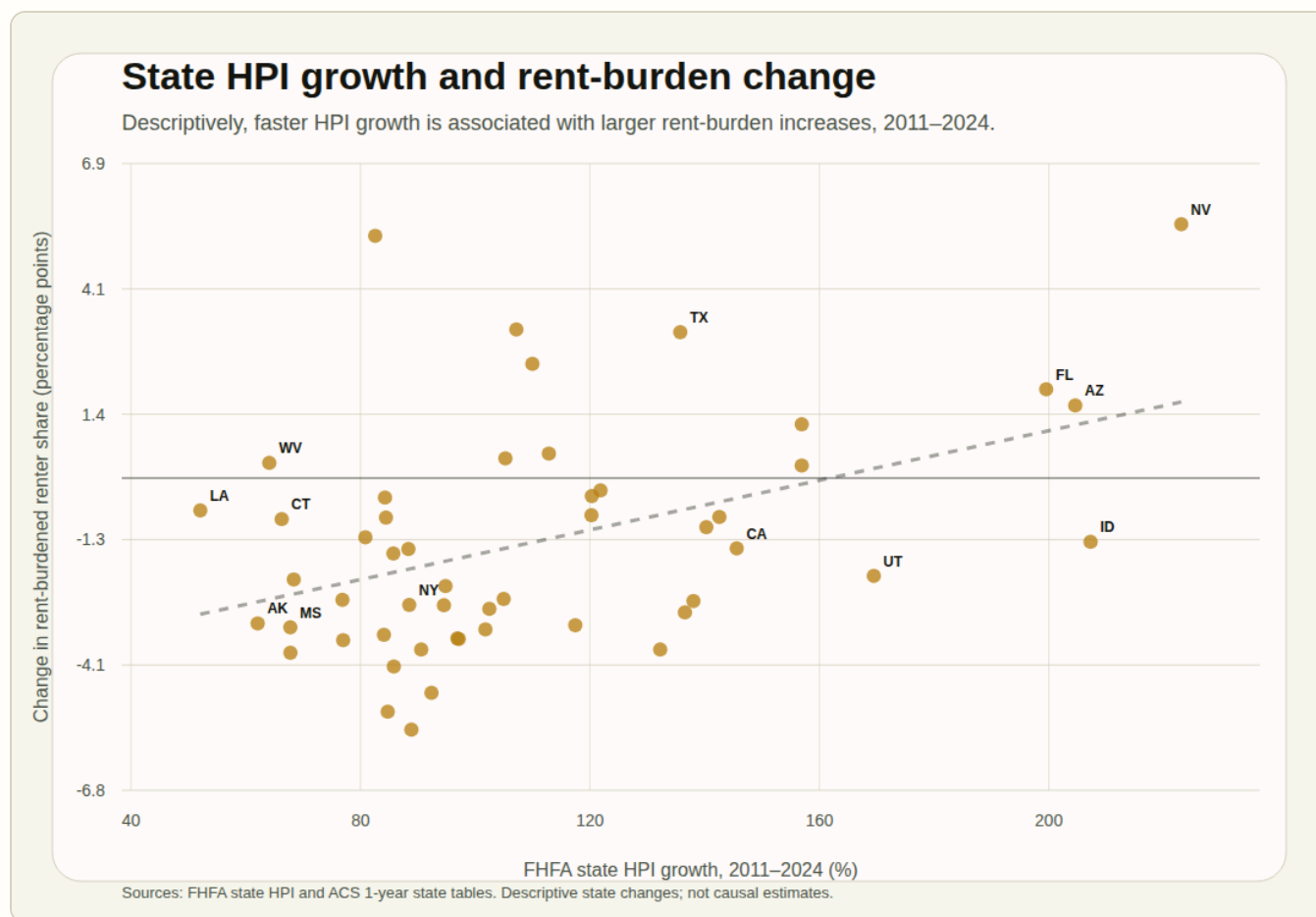


Figure 5. State HPI growth and rent-burden change

5.2 Household-formation proxies are weaker

The household-formation proxies are more tentative. HPI growth is only weakly negatively correlated with under-35 householdership ($r = -0.07$) and ownership among under-35 householders ($r = -0.10$).

That does not mean household formation is unaffected. It means these first proxies are not strong enough to carry the claim by themselves. Under-35 householdership counts young adults who are listed as householders. It misses young adults living with parents, relatives, roommates, or partners where they are not the named householder. Ownership among under-35 householders has a similar limitation: it only looks at young adults who have already formed a household.

So the local extension strengthens the rent-burden and affordability-pressure story more than the household-formation story. The household-formation channel remains plausible, but it needs better measures.

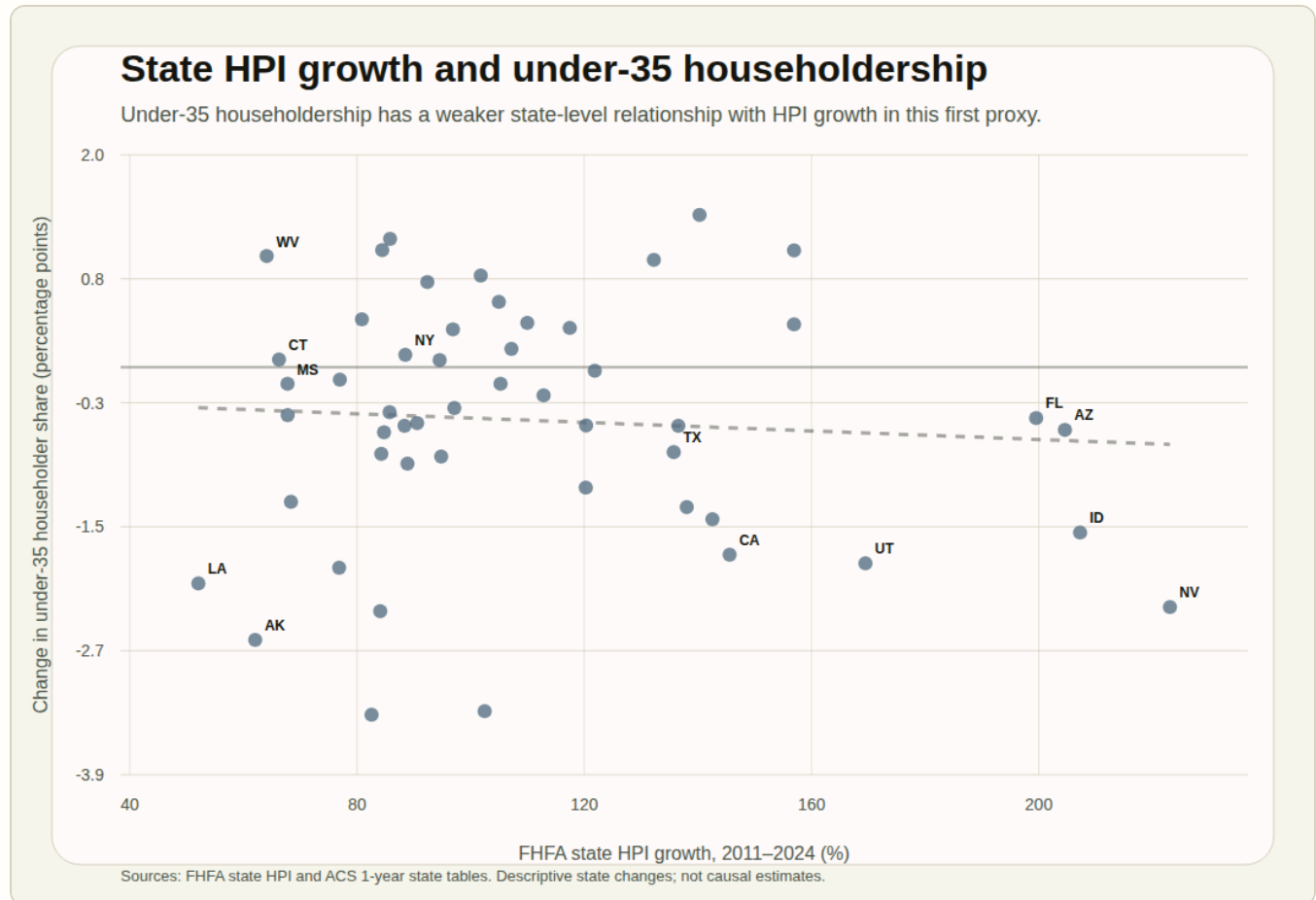


Figure 6. State HPI growth and under-35 householdership

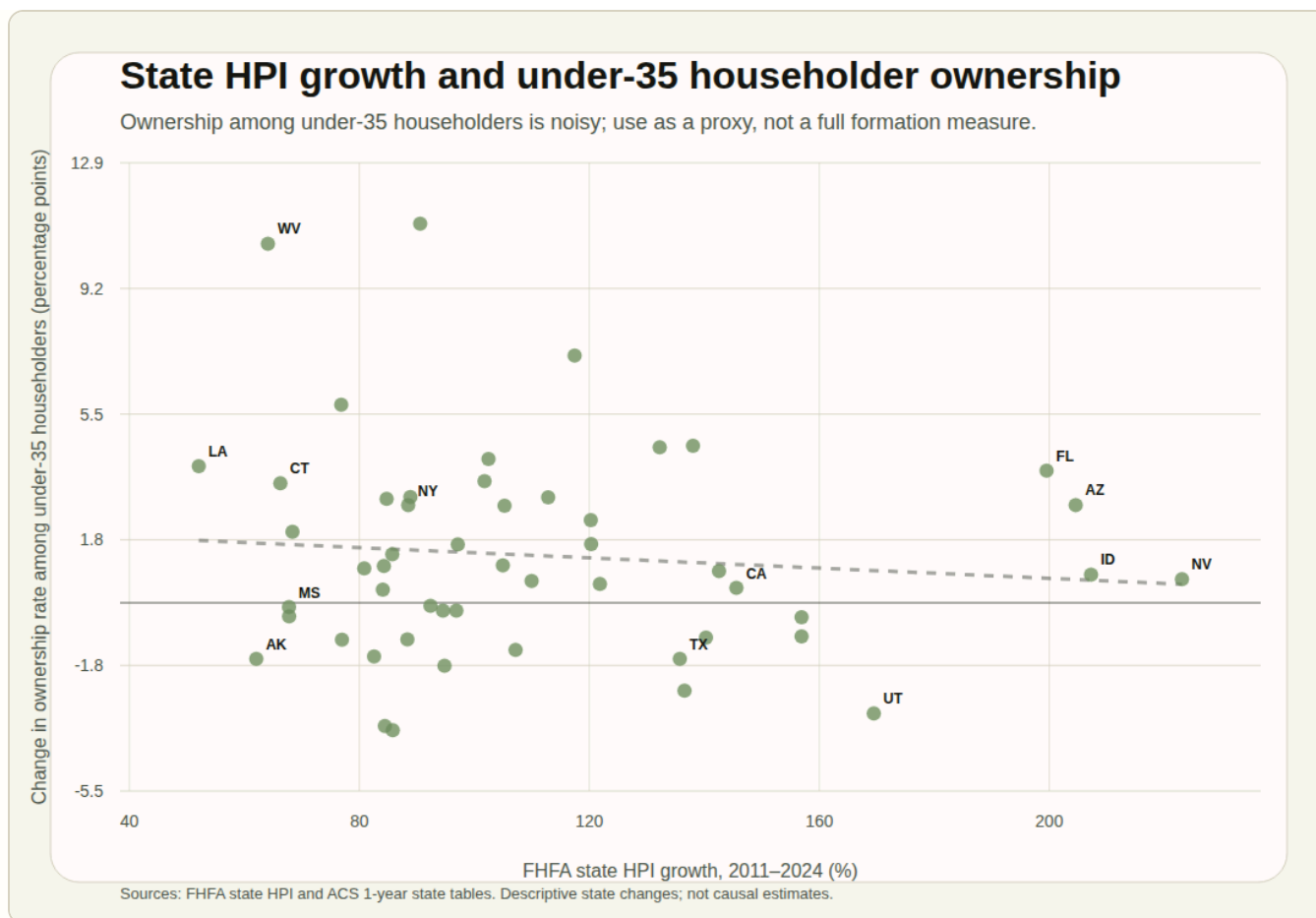


Figure 7. State HPI growth and ownership among under-35 householders

5.3 State-year panel models

Endpoint correlations can be shaped by outliers and by differences across states. To add a stronger discipline check, Table 2 estimates state-year panel models with state and year fixed effects. These models absorb stable state differences and common national year shocks. They remain descriptive associations, not causal estimates.

The panel models support the rent-burden channel. In the preferred specification, the outcome is the rent-burdened renter share and the exposure is $\log(\text{FHFA HPI divided by ACS median household income})$. With state and year fixed effects, the coefficient is 0.0772. Using state-clustered standard errors, the standard error is 0.0147 and the t-statistic is 5.25. In an alternative specification using $\log \text{FHFA HPI}$ and $\log \text{income}$ separately, the coefficient on $\log \text{HPI}$ is 0.1021 with a clustered standard error of 0.0126.

The young-householder results remain weaker. The preferred under-35 householdership specification has a coefficient of -0.0075 with a state-clustered standard error of 0.0097. Ownership among under-35 householders does not show a clear relationship with local HPI pressure.

Table 2. ACS/FHFA state-year panel models with state-clustered standard errors *All models include state and year fixed effects. Associations are descriptive, not causal.*

Outcome	Exposure	Controls	N	Coef.	Cluster SE	t	R ²
Rent-burdened renter share	log(HPI / income)	none	663	0.0772	0.0147	5.25	0.904
Rent-burdened renter share	log(HPI)	log income	663	0.1021	0.0126	8.14	0.909
Rent-burdened renter share	log(HPI)	none	663	0.0410	0.0143	2.87	0.898
Under-35 householder share	log(HPI / income)	none	663	-0.0075	0.0097	-0.78	0.967
Under-35 householder share	log(HPI)	log income	663	-0.0183	0.0092	-1.99	0.970
Under-35 householder share	log(HPI)	none	663	0.0012	0.0082	0.15	0.967
Ownership among under-35 householders	log(HPI / income)	none	663	0.0058	0.0231	0.25	0.935
Ownership among under-35 householders	log(HPI)	log income	663	-0.0072	0.0273	-0.26	0.936
Ownership among under-35 householders	log(HPI)	none	663	0.0124	0.0170	0.73	0.935

The local panel therefore sharpens the paper's hierarchy of evidence: rent burden is the strongest local finding; under-35 household formation remains exploratory.

6. Reading the Evidence as a Household

The most practical use of this paper is not to forecast home prices or declare a single cause of unaffordability. It is to separate the questions a household actually faces.

First, there is the **price hurdle**: how far local home prices have moved relative to income. Second, there is the **payment hurdle**: what mortgage rates do to the monthly obligation on a given price level. Third, there is the **cash-entry hurdle**: down payment, closing costs, reserves, taxes, insurance, and repair risk. Fourth, there is the **rent-burden channel**: whether waiting outside ownership still allows enough room to save and maintain flexibility.

Those channels can point in different directions. A low-rate period can make monthly payments look easier while the price and down-payment hurdles get worse. A high-rate period can make monthly payments worse even if price growth slows. A household with family wealth, existing equity, or unusually high savings lives in a different practical regime than a renter trying to accumulate a first down payment from wages alone.

For individual decision-making, the paper's lesson is therefore diagnostic rather than prescriptive: do not ask only whether housing is expensive. Ask which bridge is binding. Is the problem price, payment, cash entry, rent burden, income volatility, local supply, or timing risk? Better decisions start by naming the actual constraint.

7. Discussion

The evidence supports a careful version of the fiat-squeeze frame. The bridge from household income to ownership access widened, narrowed temporarily through lower rates, and widened again when high prices met higher financing costs. This is not a story of affordability getting worse in a straight line every year. It is a story of regime dependence.

That distinction matters. A low-rate environment can make monthly payments look manageable for a time, but it can also coexist with higher asset prices. When rates later rise, households that did not already own the asset face both the elevated price level and the renewed financing burden. Late entrants must cross a bridge that earlier entrants crossed under different price and rate conditions.

The local extension adds an important second layer. Rent burden is where the state-level evidence is clearest. States with faster home-price growth tended to see larger increases in rent-burdened renter shares, and the state-year panel models preserve a positive association between local housing-price pressure and rent burden after accounting for state and year fixed effects. This does not prove causality, but it makes the affordability-pressure channel harder to dismiss as only a national chart pattern.

The household-formation evidence is more preliminary. The under-35 proxies do not yet justify strong claims about delayed household formation. A stronger version should add direct measures of young adults living with parents, age-specific household headship, or microdata-based living-arrangement outcomes.

The broader implication is about personal economic agency. A household's ability to save, move, buy, form a household, and build resilience depends not only on income, but on the terms under which income converts into ownership. Monetary and credit regimes are part of those terms, but the household-facing constraint is more specific: the interaction of price, payment, cash entry, rent burden, and timing.

8. Limitations

This paper is descriptive. It cannot isolate causality. M2 is an imperfect monetary-regime measure, and the M2/CPI ratio is not a structural model. The Case-Shiller national index hides local variation. Median income hides distributional differences. The payment-to-income proxy does not include down payments, taxes, insurance, credit scores, underwriting constraints, or local price levels.

The ACS/FHFA extension improves geographic variation but remains ecological. State-level relationships are not household-level relationships. ACS margins of error are not incorporated in this draft. The panel models use state-clustered standard errors, but they are still observational models, not causal identification strategies. The under-35 householdership variables are proxies and omit young adults who are not householders.

The household-formation side is therefore the main area for improvement. Future versions should add age-specific living arrangements, young-adult headship rates with population denominators, rent burden by age group, and possibly ACS PUMS or CPS microdata.

9. Conclusion

The public data support a cautious but meaningful version of the fiat-squeeze thesis. The strongest evidence is not that every affordability measure worsened every year, or that one monetary variable explains housing outcomes. The stronger evidence is that the bridge from income to ownership became more regime-dependent.

Home prices rose faster than income over the national sample. Low mortgage rates temporarily softened the monthly-payment channel, but higher prices plus higher rates renewed the squeeze by 2024. At the state level, faster housing-price growth is most clearly associated with higher rent burden.

That matters because households need more than income. They need a reliable way to convert income into security: savings, shelter, ownership, mobility, and resilience. The practical takeaway is to break the problem into parts. A household should know whether its binding constraint is the price level, the monthly payment, the down payment, rent burden while waiting, or the risk of entering a different price-rate regime later. When that conversion from work to security becomes less reliable, the question is not only academic. It is personal: what conditions allow ordinary households to plan, save, form households, and build durable lives?

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Data transparency note

The analysis uses public data from FRED, FHFA, and the American Community Survey. The national series are annualized and indexed where needed so that price, income, payment, and monetary-regime measures can be compared over time. The state extension combines FHFA state house-price indexes with ACS 1-year state estimates for the available years from 2011 through 2024, excluding 2020 because the standard ACS 1-year release is unavailable.

The paper's empirical claims should be read as descriptive. They are intended to document patterns in public data, not to establish a full causal model of housing affordability or household formation.